

Project Report: Understanding Wireshark

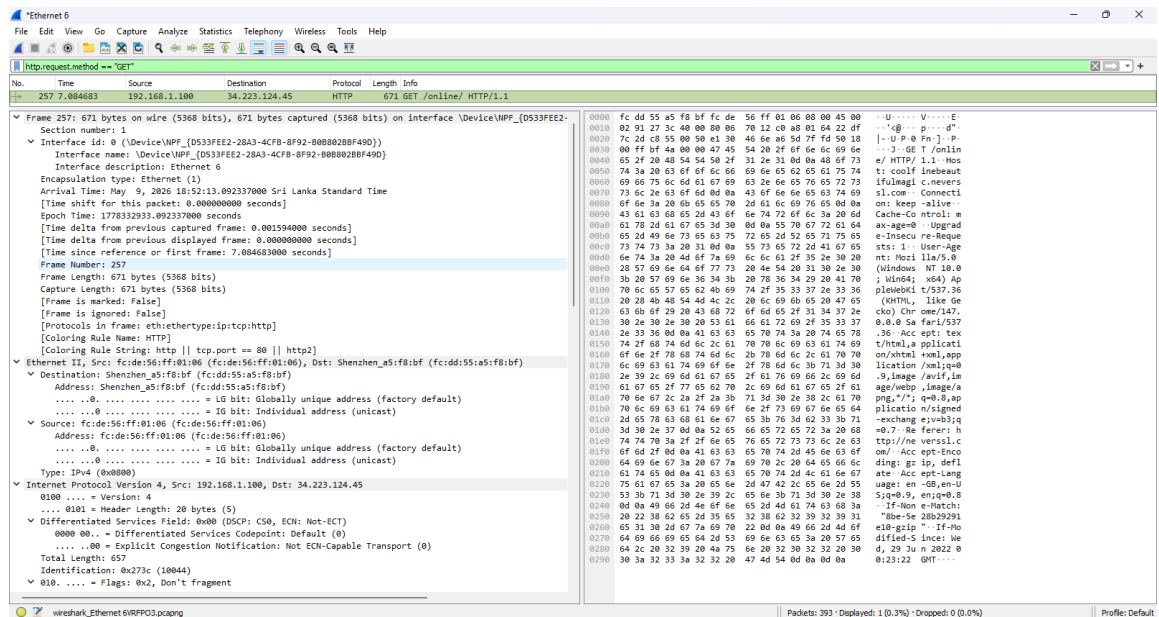
Dissecting Packets: From HTTP to TLS

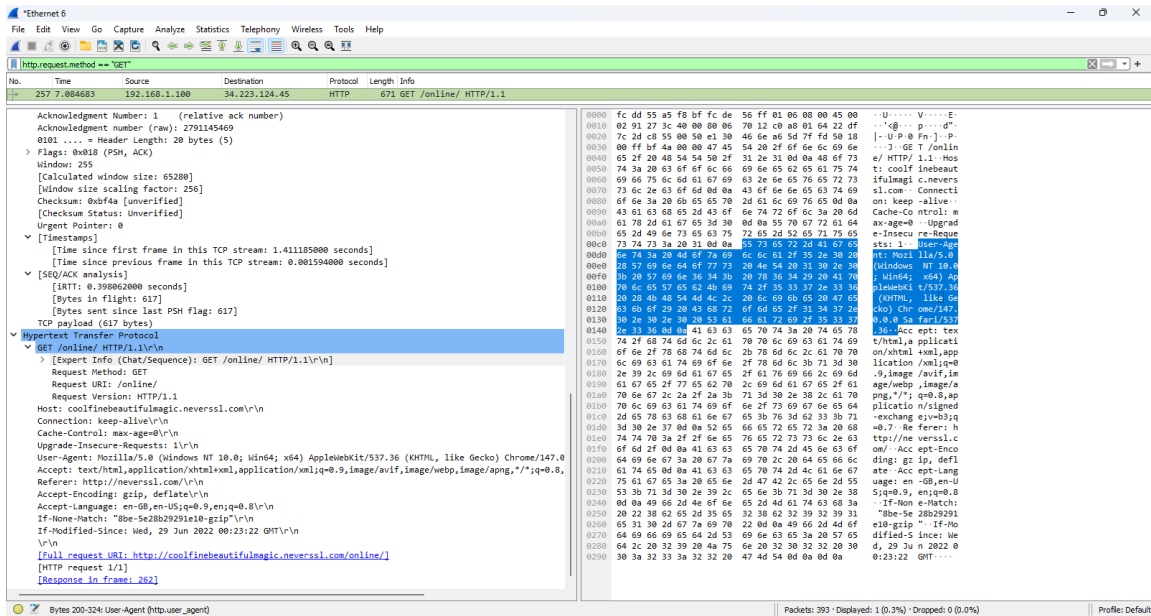
1. Introduction

This report documents the practical analysis of network traffic using Wireshark. By capturing and inspecting packets across different protocols, we can observe how data is structured, resolved, and secured over a network. The analysis covers cleartext application-layer protocols (HTTP), domain name resolution (DNS), and secure communication handshakes (TLS).

2. HTTP Traffic Analysis

The HTTP (Hypertext Transfer Protocol) traffic was filtered using the display filter `http.request.method == "GET"`. The capture demonstrates a cleartext request to a non-HTTPS web server.





Findings & Explanation:

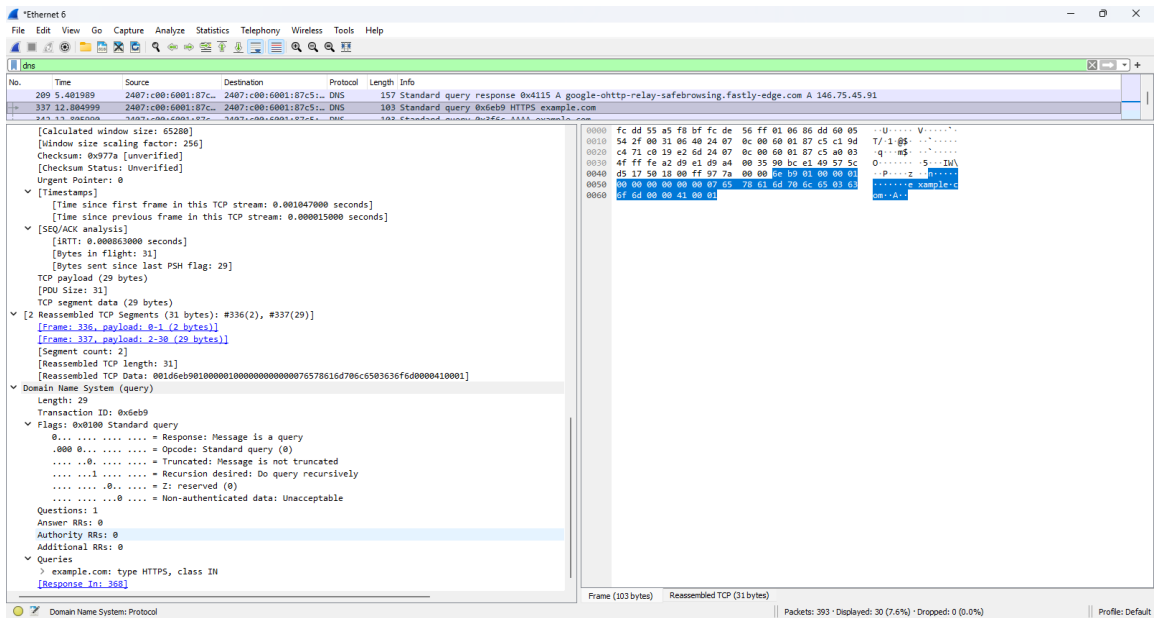
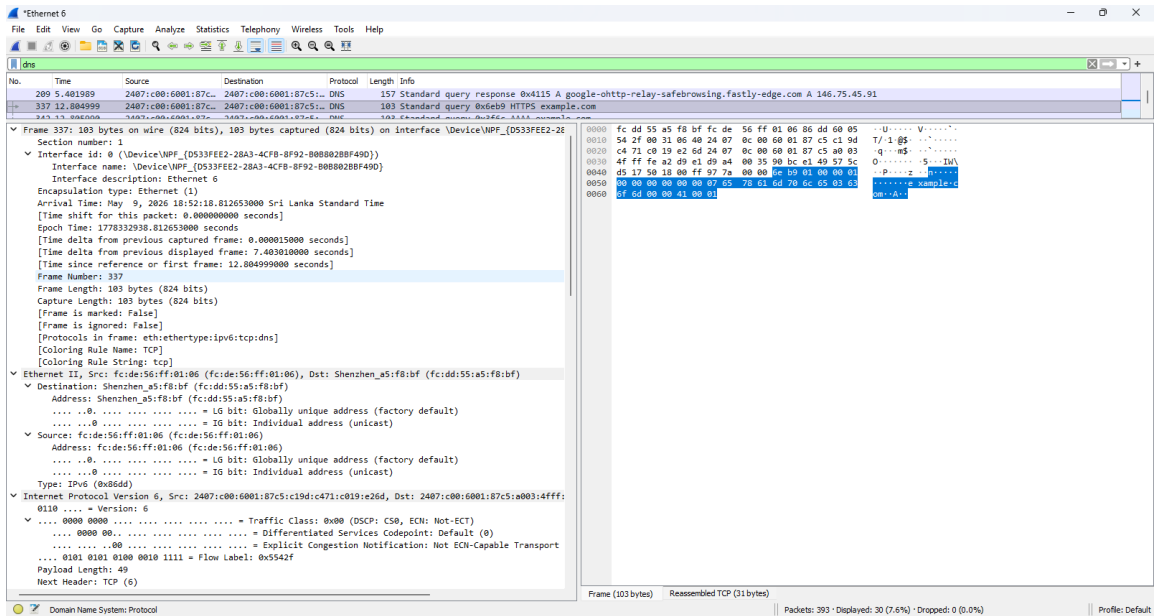
By inspecting Frame 257, we observe the following details about the connection:

- Source IP: 192.168.1.100
- Destination IP: 34.223.124.45
- Request Method: GET
- URI: /online/
- Host: coolfinebeautifulmagic.neverssl.com
- User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/147.0.0.0 Safari/537.36

Analysis: Because HTTP is an unencrypted protocol, all application layer data is transmitted in plain text. An attacker intercepting this traffic can easily read the exact URL requested, the host server, and metadata about the client's browser and operating system (User-Agent). The lack of encryption highlights the security vulnerabilities of legacy web protocols.

3. DNS Traffic Analysis

Domain Name System (DNS) traffic was isolated using the dns display filter. This capture illustrates how a domain name is resolved into an IP address prior to establishing a connection.



Findings & Explanation:

By inspecting the captured DNS frames (Frame 337), we can identify the query parameters:

- Protocol details: Queries over IPv6 network
- Transaction ID: 0x6eb9
- Queried Domain: example.com
- Query Type: HTTPS (type 65, class IN)

Analysis: The DNS query reveals that the client is attempting to securely connect to "example.com" by requesting the HTTPS binding records. The matching Transaction ID is used by the client to map the subsequent incoming DNS response to this specific outgoing

query. Analyzing DNS packets is crucial for monitoring outbound connections and identifying potential malware beacons or unauthorized communication attempts.

4. TLS Handshake Analysis

To analyze secure communications, traffic was filtered using the `tls display filter`. We captured the initial phases of a Transport Layer Security (TLS) handshake, specifically the Client Hello and Server Hello packets

The screenshot shows a Wireshark capture of a TLS handshake. The packet list on the left shows three frames: 37 (Client Hello), 56 (Change Cipher Spec), and 59 (Application Data). The packet details pane for frame 37 is expanded, showing the TLSv1.3 Record Layer and the Client Hello message structure. The message includes fields for Content Type, Version, Length, Random, Session ID, Session ID Length, Session ID, Cipher Suites, Compression Method, and Extensions. The Extensions field is expanded, showing key_share, supported_versions, signature_algorithms, and other extensions. The packet bytes pane on the right shows the raw hex and ASCII data of the packet.

The screenshot shows the fourth frame of the TLS handshake, which is a Server Hello message. The packet list on the left shows frame 60 (Server Hello). The packet details pane for frame 60 is expanded, showing the TLSv1.3 Record Layer and the Server Hello message structure. The message includes fields for Content Type, Version, Length, Random, Session ID, Session ID Length, Session ID, Cipher Suites, Compression Method, and Extensions. The Extensions field is expanded, showing key_share, supported_versions, signature_algorithms, and other extensions. The packet bytes pane on the right shows the raw hex and ASCII data of the packet.

Findings and Explanation:

Client Hello (Frame 37):

- Purpose: The client initiates the handshake, offering supported TLS versions and a list of supported Cipher Suites to the server.
- Session ID: 91a579572cb75...

Server Hello (Frame 56):

- Purpose: The server responds, officially selecting the encryption parameters that both parties will use for the session.
- Negotiated Cipher Suite: TLS_AES_128_GCM_SHA256 (0x1301)
- Version: TLS 1.2 (0x0303) in the record layer, though extensions likely step it up to TLS 1.3 parameters.

Analysis: The TLS handshake demonstrates how clients and servers establish trust and agree on encryption methods before sending application data. By successfully negotiating the TLS_AES_128_GCM_SHA256 cipher suite, the client and server ensure that all subsequent traffic in this stream will be encrypted with Advanced Encryption Standard (AES) operating in Galois/Counter Mode (GCM), ensuring both confidentiality and data integrity.

5. Conclusion

This packet analysis exercise successfully demonstrated the structural differences between cleartext and encrypted traffic flows. Capturing raw packets using Wireshark provided direct visibility into HTTP vulnerabilities, the mechanics of DNS resolution, and the cryptographic negotiation process of TLS. These fundamental analysis skills are essential for network troubleshooting, optimizing performance, and monitoring security threats.